

ISLAMIC UNIVERSITY OF TECHNOLOGY(IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)
CSE 4603: Software Testing & Quality Assurance
Quiz 4 SET B

Time 15 minutes

Marks 20

Student ID : _____

1. Which of the following is an example of an external attribute?
 - a. Total test points
 - b. Defect Density
 - c. Productivity
 - d. Program level
2. Mention two reasons for using Function Point instead of LOC as a comparable metric.
2_ Program Language dependent _____
_Some lines are different than others _____
3. Which of the following can be called primitive metrics?
 - i. LOC
 - ii. FPA
 - iii. Defect Density
 - a. i only
 - b. i & iii
 - c. ii & iii
 - d. i, ii & iii
4. Why is identifying boundaries important for calculating function points?
 - a. Including functions in the calculation
 - b. Identifying the user and external source
 - c. Define the scope of the project
 - d. All of the above
5. For a project, the function point was calculated at 271. The values for function dependent factor taken into account are: Function Importance - Normal (6); Use intensity - High (12); Interfacing - High (8); Complexity - Normal (4). Consider all dynamic characteristics as very important (5). Calculate the dynamic test point for this project.
FP=271;
 $FDC_w = ((6+12+8+4)/20)*1=1.5$;
 $QC_{dw} = (5*0.75+5*0.05+5*.1+5*0.1)/4 = 5/4=1.25$;
_____ $FP*FDC_w*QC_{dw}=271*1.5*1.25=508.125$ _____

6. Which activity does not require software measurements?

- a. Development
- b. Understanding
- c. Control
- d. Improvement

7. Mention each of the function types? Using any of these types - Internal Logical Files, External Interface Files, External Input, External Output and External Inquiries.

Search Bar _____ Inquires _____

Confirmation Message on clicking submit _____ Output _____

Store Customer order _____ Internal logical file _____

Generated Invoice _____ Output _____

Notification Messages _____ Output _____

8. Out of all 14 general characteristics of a system, ABC project only requires four of them (Data communication, Transaction rate, Online data entry and Online update) at a significant (4) level. Calculate the value adjusted factor for the mentioned project.

$$4 * 4 * 0.01 + 0.65 = 0.81$$

9. In a project, the function point was calculated at 432, and the number of bugs found was 94. Calculate the defect density of this project.

$$94 / 432 = 0.217$$

10. In a module implementation of a project, there are 21 unique operators, 15 unique operands, 89 total operators, and 38 total operands appearing in the module implementation. Calculate its Halstead Program Level.

$$PL = 1 / ((21/2) * (38/15)) = 0.0376$$